

Bayesian Market Concentration for Multi-Sector Firms

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The HHI Has a Measurement Problem

Standard HHI/CR4 treats firm-industry assignment as known.

Amazon North America (FY2024, \$386B):

- Groceries (Whole Foods + Fresh)
- General merchandise
- Consumer electronics
- Digital content + Prime

All of this gets one NAICS code. Which market is Amazon in?

When the code changes, HHI jumps—not because competition changed.

What We Know (and Don't)

Rising U.S. concentration is well-documented:

- Grullon, Larkin, Michaely (2019): 75% of industries more concentrated, 1997–2014
- Autor et al. (2020): superstar firms capture rising revenue share

But every study treats the firm-industry map as fixed.

Question

How much of measured concentration is real consolidation versus an artifact of arbitrary NAICS assignment?

This Paper

- Treat firm-industry allocation as a latent variable, identify it from Compustat NAICS co-occurrence, and propagate posterior uncertainty into concentration measures.

Main findings:

- ① Aggregate HHI rose ≈ 330 points (18%) from 2022 to 2025
- ② On average, standard benchmark **overstates** concentration but **understates** growth over time
- ③ Benchmark 2025 HHI lies in **right tail** of Bayesian posterior

Data: WRDS API Compustat Segment Panel

Source: Compustat wrds_segmerged, 2017–2025

Sales panel

- 19,302 segment-year obs
- 2,932 unique firms
- 241 current S&P 500 members
- S&P 500 = 60% of total sales
- S&P 500 median: \$6.8B/segment

NAICS count panel

- 41,844 segment-industry rows
- 862 unique NAICS codes
- Mean count per row: 2.91
- Max: 12 NAICS codes/segment
- Counts $\rightarrow$ Dirichlet prior α_0

Stable coverage 2019–2022 ($\approx 2,200$ firms/yr). Partial-year attrition in 2024–2025.

Model: Bayesian Dirichlet Allocation

Indices: $f = \text{firm}$, $s = \text{segment} \in f$, $k = \text{industry}$, $t = \text{year}$, $r = \text{draw}$

Prior Sales-weighted pseudo-counts (summed over pre-period $\mathcal{T}_0 = 2017\text{--}2021$):

$$\alpha_{fs,k}^{(0)} = \sum_{t \in \mathcal{T}_0} \underbrace{\frac{c_{fs,k,t}}{C_{k,t}} \cdot S_{k,t}}_{w_{fs,k,t}: \text{segment's share of industry } k\text{'s count} \times \text{industry sales}} + \varepsilon$$

Closed-form posterior for segment (f, s) in year t :

$$\pi_{fs,\cdot,t} \mid \text{data} \sim \text{Dirichlet}(\alpha_{fs,\cdot}^{(0)} + w_{fs,\cdot,t})$$

Posterior draws ($R = 500$) are taken from this known distribution only to propagate uncertainty into HHI and CR4:

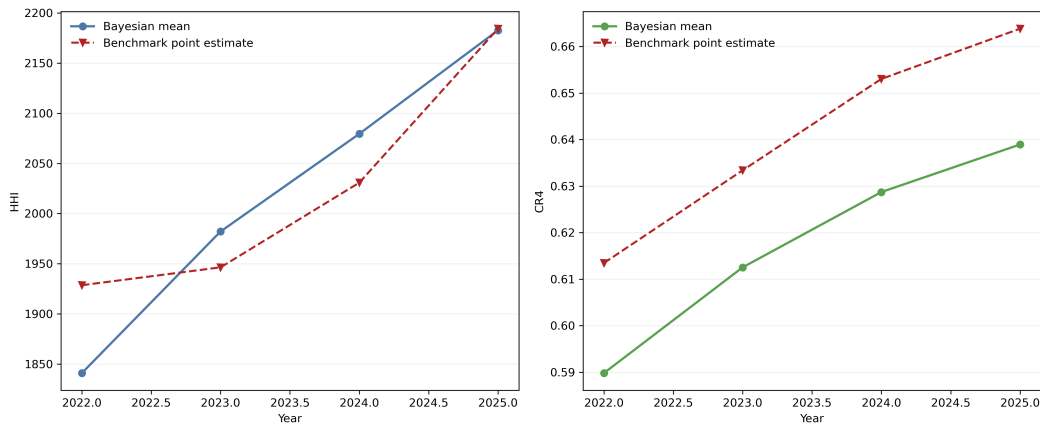
$$\pi_{fs,\cdot}^{(r)} \sim \text{Dirichlet}(\alpha_{fs,\cdot}^{(0)} + w_{fs,\cdot,t}), \quad \pi_{fs,k}^{(r)} = \text{drawn share of } (f, s)\text{'s revenue in } k$$

Firm imputed revenue in industry k , year t , draw r :

$$\tilde{r}_{f,k,t}^{(r)} = \sum_{s \in f} \pi_{fs,k}^{(r)} \cdot r_{fs,t} \Rightarrow \text{market shares} \rightarrow \text{HHI, CR4.}$$

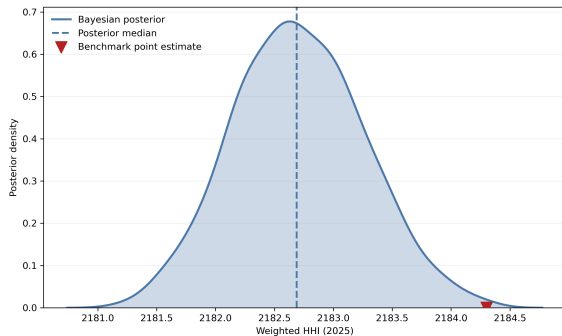
Benchmark (no draws): fixed $\hat{\pi}_{fs,k,t} = c_{fs,k,t} / \sum_{k'} c_{fs,k',t}$ (raw count, normalized in year t)

Aggregate HHI Rose 330 Points, 2022–2025



Solid (circles): Bayesian posterior mean. Dashed (triangles): benchmark. **Left:** HHI 1,850→2,180. **Right:** CR4 0.59→0.64.

Benchmark Overstates 2025 Concentration



Finding

Benchmark ($\approx 2,184.5$, triangle) lies **right of** the posterior median ($\approx 2,182$ – $2,183$, dashed line).

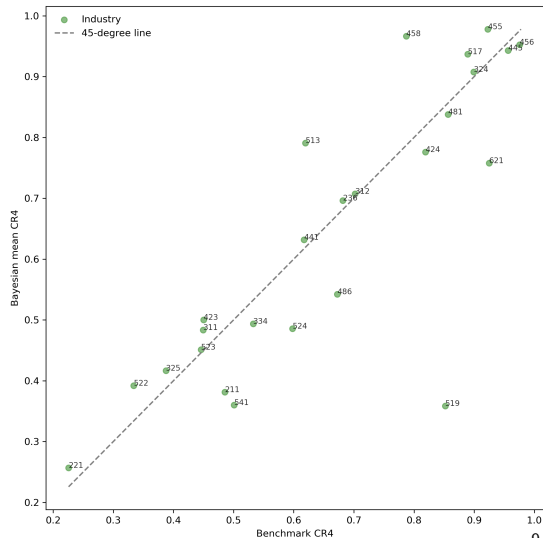
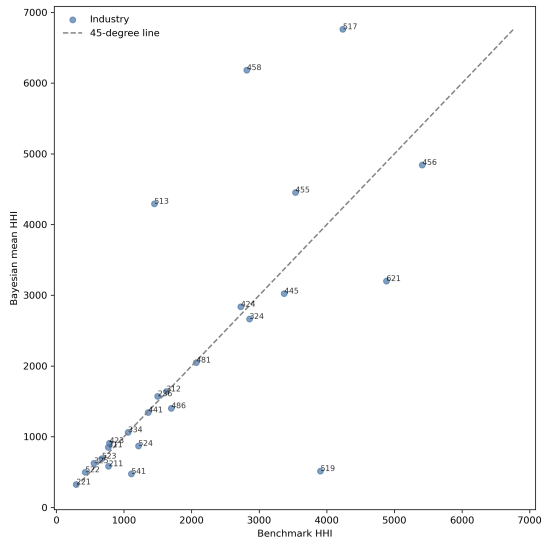
Why? Fixed weights place revenue in modal NAICS code

- over-concentrates sales for leading firms
- inflates HHI and CR4 systematically

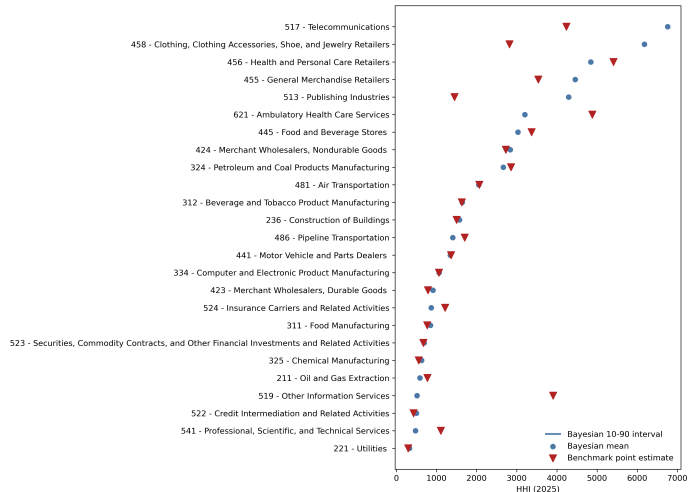
Posterior is tight (± 1.5 HHI units)

- gap is bias, not simulation noise

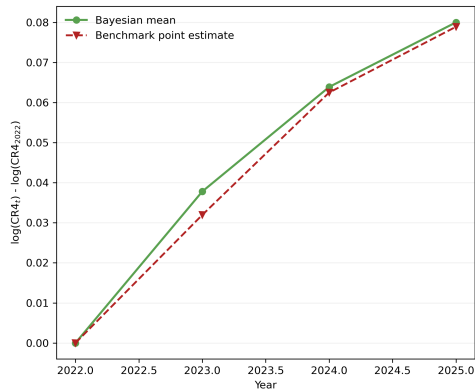
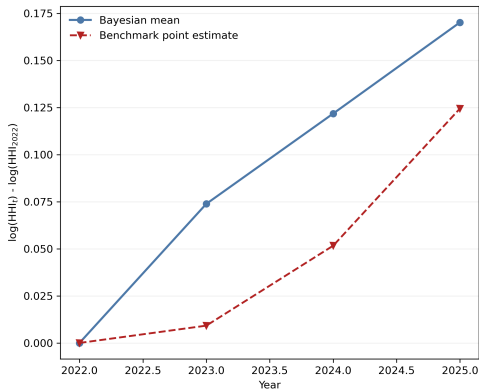
Benchmark Overstates Across Industries (2025)



Cross-Industry Heterogeneity in Posterior Uncertainty



Concentration Growth Is Robust



Log-change from 2022. Lower bound of 2025 credible band is above zero in every post-2022 year. Trend is structural.

NAICS 3-Digit Evidence

Finding 1: CR4 is more distorted than HHI

Traditional HHI point estimates fall closer to—and more often within—the Bayesian HHI posterior than traditional CR4 estimates do to the Bayesian CR4 posterior.

Why: CR4 is a rank statistic. Small allocation shifts can reshuffle the top-four; HHI is a smooth quadratic—more robust to rank permutations.

Finding 2: Benchmark overstates levels but understates growth

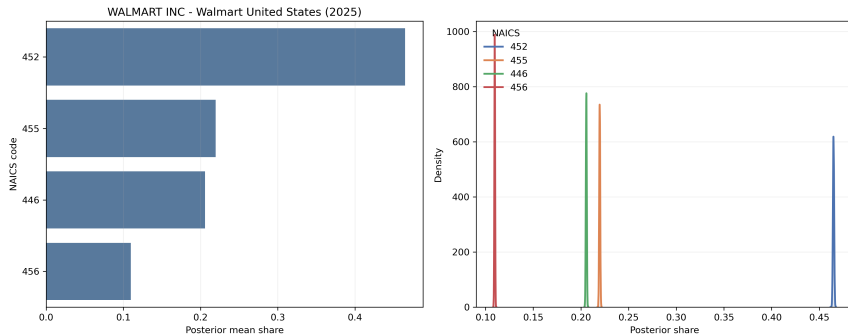
Fixed-assignment measures overstate concentration **today** and understate how fast it is rising.

Why: In the data, the Bayesian series starts from a lower baseline and rises faster. One interpretation is that ambiguous segment revenue is initially allocated more diffusely and then shifts toward dominant firms as later-year evidence accumulates. The deterministic benchmark collapses each year to one fixed allocation vector, so it misses this intertemporal uncertainty margin.

Takeaways

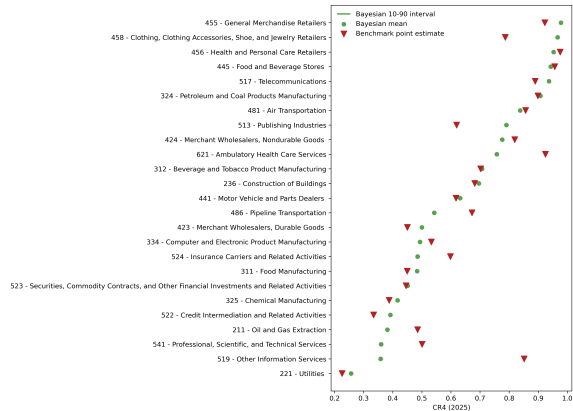
- ➊ **Concentration has risen sharply post COVID.** Sales-weighted HHI up 330 points (18%) from 2022 to 2025.
- ➋ **Standard measures overstate concentration levels.** Benchmark sits in the right tail of the Bayesian posterior—systematic bias, not noise.
- ➌ **...but understate concentration growth.** Fixed yearly assignment misses the steeper trend the Bayesian model reveals once allocation uncertainty is propagated over time.
- ➍ **CR4 is more distorted than HHI.** Rank-order sensitivity makes CR4 more susceptible to small allocation shifts.
- ➎ **Both errors matter for measuring competition.** Over-intervention today, under-intervention for growth over time.

Appendix: Segment-Level Dirichlet Posterior



Posterior allocation for a single large 2025 segment with ≥ 3 NAICS links. Meaningful within-segment uncertainty even with an informative prior.

Appendix: CR4 by Industry, 2025



Benchmark-posterior gap larger for CR4 than for HHI—fixed allocation inflates top-four share more aggressively than aggregate HHI.

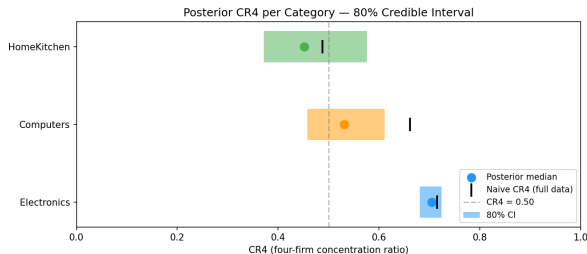
Appendix: Coverage by Year

Statistic	Firms	Current S&P 500 firms	Segment-year observations	Segment-industry rows	Total sales	Total count mass
2017	1,728	168	1,797	3,698	3,930,581.97	5,397.00
2018	2,037	197	2,139	4,472	5,739,931.17	11,463.00
2019	2,238	205	2,365	4,990	6,565,124.91	18,042.00
2020	2,228	207	2,347	5,457	6,436,082.00	18,312.00
2021	2,228	207	2,355	5,636	7,196,233.59	18,171.00
2022	2,209	206	2,329	5,119	7,954,570.72	17,943.00
2023	2,148	207	2,261	4,795	8,248,851.77	16,542.00
2024	2,024	203	2,102	4,365	8,260,132.23	10,818.00
2025	1,508	197	1,571	3,240	8,116,667.43	4,821.00
2026	34	11	36	72	1,267,646.34	108.00
All years	2,932	241	19,302	41,844	63,715,822.11	121,617.00

Appendix: Model Generalizes — Amazon India (Kaggle)

Same model, different data.

- 1,465 Amazon India product listings; revenue $\propto$ price $\times$ rating count
- Brands $\leftrightarrow$ firm-segments Categories $\leftrightarrow$ NAICS codes
- Keyword co-occurrence counts $\leftrightarrow$ NAICS linkage counts
- 20% evidence sample with revealed categories updates posterior



Same pattern reappears:

Naïve benchmark overstates CR4

Largest gap in **Computers**:

– Bayesian: 0.53 [0.46, 0.61]

– Naïve: 0.66 (outside CI)

Electronics: near-agreement
(allocation uncertainty is lower)